

WHISPERS FROM MARS

Stage 3 Technical Submission

Section 8 — Executive Scientific & Mission Verdict

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Target Outcrop: Cheyava Falls, Neretva Vallis, Jezero Crater

Core Sample: #22 “Sapphire Canyon” — Tube 25, Sol 1215

Introduction: The Search for Ancient Life in Jezero Crater

For over half a century, Mars has stood as the most tantalizing candidate for life beyond Earth in our solar system. Once a colder, drier world stripped of the thick atmosphere and global magnetic field it may have had in its youth, Mars nonetheless preserves an extraordinary geological record: dried river deltas, mineral-rich lakebeds, and networks of ancient channels that speak of a planet that was once warmer and wetter. Jezero Crater, the landing site of NASA's Perseverance rover since February 2021, was selected precisely because it once hosted a lake fed by an inflowing river delta — a setting on Earth where organic material and microbial signatures are readily preserved in fine-grained sediment.

Neretva Vallis, the breach channel that once carried water into Jezero, and its surrounding delta margins have become the focus of Perseverance's astrobiological campaign. It was here, at an outcrop named Cheyava Falls, that the rover's arm-mounted instruments — SHERLOC, PIXL, and the WATSON microscopic imager — detected an arresting pattern: leopard-spot-like reaction fronts, organic carbon signatures, and iron-mineral haloes embedded in a vein-laced mudstone. Such a combination is exactly the kind of biosignature-adjacent feature the mission was designed to hunt for, and exactly the kind of feature that also has plausible, well-documented abiotic explanations rooted in hydrothermal and diagenetic chemistry.

This submission presents the Stage 3 technical verdict for Core Sample #22 ("Sapphire Canyon," Tube 25, Sol 1215), drawn from the Cheyava Falls outcrop. Using quantitative isotopic, stoichiometric, thermodynamic, and spatial-statistical telemetry gathered by Perseverance's instrument suite, we evaluate whether the observed features are best explained by biological activity or by abiotic geochemical processes, and we lay out the follow-on laboratory tests, missing telemetry, and rover operations needed to resolve the question with sample-return data.

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Q1 Primary Verdict — Life vs. Abiotic

OFFICIAL VERDICT: (C) Abiotic Geochemical Process

Quantitative Evidence & Justification

- **Carbon Isotope Telemetry ($\delta^{13}\text{C}$):** Measured organic $\delta^{13}\text{C} = -3.8 \pm 0.6 \text{ ‰}$ (VPDB). Terrestrial biological enzymatic fractionation falls between -25.0 ‰ and -35.0 ‰ . A value of -3.8 ‰ instead matches abiotic Martian mantle CO_2 (-2.0 to -6.0 ‰), rejecting biological carbon fixation at $>25\sigma$ confidence.
- **Sulfur & Gas Stoichiometric Baselines:** Sulfide $\delta^{34}\text{S} = +4.2 \pm 0.5 \text{ ‰}$ matches abiotic magmatic sulfur ($+2.0$ to $+5.0 \text{ ‰}$), ruling out microbial dissimilatory sulfate reduction (-15.0 to -40.0 ‰). The hydrocarbon ratio $\text{C}_1/(\text{C}_2+\text{C}_3) = 14.2 \pm 1.8$ confirms Fischer-Tropsch-Type (FTT) thermal synthesis (<50), rejecting a biogenic methane signature ($>1,000$).
- **Thermodynamic Constraints:** Pristine forsteritic olivine (Fo76) preserved in vein inclusion Pt-105 ($\text{Mg}:\text{Fe} = 3.15 \pm 0.12$) dissolves rapidly in neutral water (<100 years at 25°C). Its survival indicates high-temperature ($>200^\circ\text{C}$) serpentinization / hydrothermal alteration, not a habitable ambient surface lake.

Summary Data & Evidence Table

Telemetry / Feature Stream	Observed Telemetry	Biological Threshold	Abiotic Baseline	Scientific Verdict
$\delta^{13}\text{C}$ (Organic C)	$-3.8 \pm 0.6 \text{ ‰}$	-25 to -35 ‰	-2.0 to -6.0 ‰	Abiotic Mantle CO_2 ($>25\sigma$ rejection)
$\delta^{34}\text{S}$ (Sulfide)	$+4.2 \pm 0.5 \text{ ‰}$	-15 to -40 ‰	+2.0 to +5.0 ‰	Abiotic Magmatic Baseline
$\text{C1}/(\text{C2}+\text{C3})$ Ratio	14.2 ± 1.8	$>1,000$	<50	Abiotic Fischer-Tropsch Synthesis
Spatial Dispersion (R)	$R = 1.42$ ($p < 0.01$)	$R < 1.0$ (Clustered)	$R > 1.0$ (Dispersed)	Abiotic Chemical Diffusion Front
Olivine (Fo76) Inclusions	Pristine Mg:Fe = 3.15	Dissolves in <100 yrs	Stable $>200^\circ\text{C}$	High-Temp Hydrothermal Activity
Steno's Law Timeline	Vein A truncates Pt-102	N/A	Cross-cutting vein	Post-halo secondary fracturing

Q2 Plausible Non-Biological Pathway

Abiotic low-temperature Fischer-Tropsch-Type (FTT) hydrothermal catalysis on basaltic mineral substrates synthesized polycyclic aromatic hydrocarbons (1598 cm^{-1} G-band at Pt-102). Subsequent abiotic redox reactions between mantle-derived organic reductants and magmatic Fe^{3+} in the mudstone precipitated vivianite (Fe-P, Pt-102) dark halo rims and greigite (Fe-S, Pt-101) bleached cores via self-limiting chemical diffusion ($R = 1.42$), forming pseudo-biogenic “leopard spots” without living organisms.

Q3 Falsification & Earth Laboratory Test

Upon return of Sample Tube #22 to Earth, perform Compound-Specific Isotope Analysis (CSIA) via Gas Chromatography-Isotope Ratio Mass Spectrometry (GC-IRMS) and synchrotron nano-XRD to measure homochirality / enantiomeric excess in amino acids and organic molecules. Demonstrating racemic (50:50 L/D) amino acid mixtures and un-fractionated carbon chains will conclusively falsify a biological origin and confirm abiotic FTT synthesis.

Q4 Critical Missing Telemetry Factor

The single biggest missing factor is in-situ paleofluid pH, temperature, and dissolved oxidation state trapped inside fluid inclusions within the vein and mudstone matrices. Rover instruments (PIXL/SHERLOC) cannot directly measure fluid inclusion thermodynamics or trapped volatile isotopic composition.

Q5 Strategic Rover Operations Command

Cache Tube #22 (“Sapphire Canyon”) and drive onward toward the Jezero Crater western margin. Do not expend further drill time or abrasion tools on Cheyava Falls. Proceed to explore older Noachian basement strata to maximize astrobiological discovery potential.